Orbit-Balanced Leaves on the Diamond Graph: Fractional Degree Bias Beyond Vertex Transitivity

Abstract

Let $D = K_4 - e$ be the diamond graph. Its automorphism group has two vertex orbits of size two: the endpoints of the shared triangle edge and the two page vertices. Attach a total of k pendant leaves within each orbit, with arbitrary distribution inside the orbit. For $1 \le k \le 2$, we prove the chromatic-polynomial graphon lower bound throughout $p \ge 1/2$. Orbit symmetrization makes the four root exponents uniformly $k/2$; a change of measure, fractional Hölder inequality, the accepted pure-chordal diamond bound, and an elementary scalar monotonicity then give the exact target. The case $k = 1$ proves the formerly open NetworkX Atlas graph 113.

1 Family, orbit structure, and target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. Put

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y).$$

Label the diamond D by spine vertices a, b and page vertices c, d, with edge set

$$ab,\ ac,\ ad,\ bc,\ bd.$$

Thus D is the union of the triangles abc and abd along the spine edge ab. Its automorphism group contains the independent swaps $a \leftrightarrow b$ and $c \leftrightarrow d$; these are exactly its two vertex orbits.

Choose nonnegative integers

$$s_a + s_b = k, \qquad q_c + q_d = k, \qquad 1 \le k \le 2,$$

and attach the indicated number of private pendant leaves to each core vertex. Denote the resulting graph by

$$D_k(s_a, s_b; q_c, q_d).$$

There are $2k$ leaves in total. Since

$$\chi_D(x) = x(x-1)(x-2)^2,$$

independent colouring of the leaves gives

$$\chi_{D_k}(x) = x(x-1)^{2k+1}(x-2)^2, \qquad \chi(D_k) = 3. \tag{1}$$

Consequently

$$\Phi_{D_k}(p) = p^{2k+1}(2p-1)^2. \tag{2}$$

This is a polynomial identity, including the continued value one at $p = 1$; at the required threshold $p = 1/2$, the target is zero.

We use the accepted pure-chordal theorem only in its diamond instance: every graphon V of edge density $s \geq 1/2$ satisfies

$$t(D, V) \geq \phi(s), \qquad \phi(s) := s(2s-1)^2. \tag{3}$$

Indeed, the two maximal cliques of D are triangles, so D is 3-pure chordal. The theorem applies on arbitrary probability spaces.

2 Fractional edge and scalar transfer

Lemma 2.1 (Fractionally degree-weighted edge)**.** *For $0 < \alpha \leq 1$, set*

$$N_\alpha = \int_{\Omega^2} W(x,y) d(x)^\alpha d(y)^\alpha \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y).$$

Then

$$N_\alpha \geq p^{1+2\alpha}.$$

Proof. Define $W(x,y)/d(x)$ to be zero on $\{d(x) = 0\}$; Fubini implies that $W(x,y)$ vanishes for almost every y there. With

$$\theta = \frac{1}{1+2\alpha}, \qquad \eta = \frac{\alpha}{1+2\alpha},$$

one has $\theta + 2\eta = 1$ and

$$W = \left(W d(x)^\alpha d(y)^\alpha\right)^\theta \left(\frac{W}{d(x)}\right)^\eta \left(\frac{W}{d(y)}\right)^\eta$$

almost everywhere on the support of W. Weighted Hölder yields

$$p \leq N_\alpha^\theta \left(\int \frac{W}{d(x)} \, \mathrm{d}\mu^2\right)^\eta \left(\int \frac{W}{d(y)} \, \mathrm{d}\mu^2\right)^\eta.$$

The last two integrals are at most one, so $p \leq N_\alpha^\theta$. □

Lemma 2.2 (Diamond rescaling)**.** *Let $k \in \{1, 2\}$, let $p \in (1/2, 1]$, and suppose $0 < z \leq p^{2k}$. Put*

$$s_0 = p \left(\frac{p^{2k}}{z}\right)^{1/2}.$$

If $s_0 \leq 1$, then

$$z\phi(s_0) \geq p^{2k}\phi(p).$$

Proof. For $s \in [1/2, 1]$, put

$$\psi(s) = \frac{\phi(s)}{s^2} = \frac{(2s-1)^2}{s} = 4s - 4 + \frac{1}{s}.$$

Then

$$\psi'(s) = 4 - \frac{1}{s^2} \geq 0.$$

The definition of s_0 is equivalent to

$$z = p^{2k}\left(\frac{p}{s_0}\right)^2.$$

Since $s_0 \geq p$, monotonicity of ψ gives

$$z\phi(s_0) = p^{2k+2}\psi(s_0) \geq p^{2k+2}\psi(p) = p^{2k}\phi(p).$$

□

The function ϕ is nondecreasing on $[1/2, 1]$, because

$$\phi'(s) = (2s-1)(6s-1) \geq 0.$$

3 The orbit-balanced theorem

Theorem 3.1 (Orbit-balanced diamond leaves)**.** *For every $k \in \{1, 2\}$, every admissible distribution $s_a + s_b = q_c + q_d = k$, and every graphon W of edge density $p \in [1/2, 1]$,*

$$\boxed{t\big(D_k(s_a, s_b; q_c, q_d), W\big) \geq p^{2k+1}(2p-1)^2 = \Phi_{D_k}(p).}$$

Proof. At $p = 1/2$ the target is zero. At $p = 1$, one has $W = 1$ almost everywhere and both sides equal one. Hence assume $p \in (1/2, 1)$.

For core variables x_a, x_b, x_c, x_d, write

$$F_D = W(x_a, x_b)W(x_a, x_c)W(x_a, x_d)W(x_b, x_c)W(x_b, x_d).$$

After integrating the leaves, the density is

$$\int F_D d(x_a)^{s_a} d(x_b)^{s_b} d(x_c)^{q_c} d(x_d)^{q_d}\, \mathrm{d}\mu^4.$$

The independent swaps within the spine and page orbits give four equal integrals. Average them and apply the arithmetic–geometric mean inequality pointwise. Each core variable then receives the average exponent $k/2$:

$$t(D_k, W) \geq \int F_D \prod_{v \in \{a,b,c,d\}} d(x_v)^{k/2}\, \mathrm{d}\mu^4. \tag{4}$$

Set

$$\alpha = \frac{k}{2} \in (0, 1], \qquad M = \int_\Omega d(x)^\alpha\, \mathrm{d}\mu(x), \qquad z = M^4.$$

Since $p > 0$, one has $M > 0$. Define the probability measure

$$\mathrm{d}\nu(x) = \frac{d(x)^\alpha}{M}\, \mathrm{d}\mu(x).$$

The right side of (4) is

$$z\, t(D, W; \nu).$$

The edge density of W on (Ω, ν) is

$$s = \frac{N_\alpha}{M^2}.$$

Concavity of u^α gives

$$M \le p^\alpha, \qquad 0 < z \le p^{4\alpha} = p^{2k}.$$

Lemma 2.1 gives

$$s \ge \frac{p^{1+k}}{M^2} = p\left(\frac{p^{2k}}{z}\right)^{1/2} =: s_0.$$

Because $W \le 1$ and ν is a probability measure, $s \le 1$, hence $s_0 \le 1$. Also $s \ge s_0 \ge p > 1/2$.

Apply the accepted diamond bound (3) on the biased space, monotonicity of ϕ, and Lemma 2.2:

$$\begin{aligned} t(D_k, W) &\ge z\, t(D, W; \nu) \\ &\ge z\phi(s) \ge z\phi(s_0) \ge p^{2k}\phi(p) = p^{2k+1}(2p-1)^2. \end{aligned}$$

This equals (2). □

At every balanced complete ℓ-partite graphon, $\ell \ge 2$, the degree is constant, the orbit symmetrization and measure-change inequalities are equalities, and the accepted diamond bound is sharp. Thus the theorem has all expected chromatic equality examples, including the zero target at $\ell = 2$.

4 Atlas 113, formalization audit, and limitations

Corollary 4.1 (Atlas 113)**.** *NetworkX Atlas graph* 113*, with graph6 code* `E^_O`*, is positive. It has order six, size seven, chromatic number three, and*

$$\chi_{F_{113}}(x) = x(x-1)^3(x-2)^2.$$

Every graphon of edge density $p \in [1/2, 1]$ *satisfies*

$$t(F_{113}, W) \ge p^3(2p-1)^2 = \Phi_{F_{113}}(p).$$

Proof. Its NetworkX edge list is

$$02,\ 03,\ 04,\ 12,\ 13,\ 23,\ 35.$$

Vertices $0, 1, 2, 3$ induce the diamond with spine $2, 3$ and pages $0, 1$. Vertex 4 is a leaf at page 0, while vertex 5 is a leaf at spine vertex 3. Hence this is the case $k = 1$ of Theorem 3.1. □

All integrands are bounded, nonnegative, and measurable, so Tonelli–Fubini applies directly. The quotient on $\{d = 0\}$ and positivity of the biased measure's normalizing constant are handled explicitly. No regularity, minimum-degree assumption, finite-step reduction, injective-copy interpretation, or numerical estimate is used. The only accepted graphon input is the pure-chordal diamond bound with exactly matching hypotheses.

The orbit-balance condition is essential to this proof: the total number of leaves on the spine orbit must equal the total on the page orbit, so that all four core variables acquire one common exponent. The restriction $k \le 2$ is exactly what makes that exponent at most one and permits the concavity step. The theorem does not cover a leaf distribution concentrated in only one orbit, a two-edge tail, or attachments to an arbitrary pure-chordal core.

The finite identities, symmetrized exponents, scalar derivative, and Atlas identification were reconstructed in exact rational arithmetic. For formal verification, the new reusable inputs are the two-orbit arithmetic–geometric symmetrization and the scalar transfer for $s(2s-1)^2/s^2$. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph113.lean`.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/CliqueDist/Diamond.lean`, ending at `satisfiesLowerBound_113`. It proves the case $k = 1$ and only that case, and it departs from this note at the same two points as the companion `notes/clique_distributed_leaves.pdf` does.

Scope: $k = 1$. The theorem here covers $1 \le k \le 2$. The formalization fixes $k = 1$, where the symmetrized exponent $k/2$ is $1/2$ and the biased density is $\sqrt{d}$; at $k = 2$ it would be 1, a different bias, and no six-vertex Atlas row needs it.

The orbit average becomes a single exchange. This note averages over both orbits, producing four monomials $d_a d_b$ with $a \in \{0, 1\}$, $b \in \{2, 3\}$, and applies AM–GM to all four. The formalization uses one: swapping $0 \leftrightarrow 1$ and $2 \leftrightarrow 3$ simultaneously is an automorphism of D, so the graph with leaves at $\{0, 2\}$ is isomorphic to the one with leaves at $\{1, 3\}$, and

$$2\,t(H, W) = \int F\big(d_0 d_2 + d_1 d_3\big) \ge \int F \cdot 2\sqrt{d_0 d_1 d_2 d_3},$$

the two-term AM–GM. The four-term average and the two-term one give the same weight $\prod_i d_i^{1/2}$; the two-term version needs one graph isomorphism (`CliqueDist.isoOrbit`, checked by `decide`) instead of four peelings.

The scalar transfer becomes a factorization. Lemma 2.2 is proved here by differentiating $A_D(s)/s^2$. The formalization instead uses, with $m = M^2$ and s the biased edge density,

$$p(2s - 1)^2 - s(2p - 1)^2 = (s - p)(4ps - 1), \tag{5}$$

a `ring` identity whose two factors are nonnegative because $s \ge p \ge 1/2$. Together with $m^2 s^2 \ge p^4$ this gives $m^2 A_D(s) \ge p^2 A_D(p)$ directly.

Everything else is reused verbatim. Lemma 2.1 at $\alpha = 1/2$ is `GeometricMean.sq_le_integral_geoIntegrand` $M \le p^{1/2}$ is $M^2 \le p$, one Cauchy–Schwarz against the constant 1; the change of measure is `CliqueDist.integral_sqrtDegree_prod`. The accepted diamond input is not re-derived: it is `BaseCone.base_diamond`, already verified for `notes/six_verified_base_cones.pdf`, which is exactly $t(D, V) \ge z(2z - 1)^2$ on $z \ge 1/2$.